

MATH3026/4022 Lecture 11

Forecasting I

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- ▶ What is forecasting?
- ▶ Mean square prediction error
- ▶ Forecasting for ARMA processes
- ▶ Examples
 - ▶ $AR(2)$ process
 - ▶ $AR(1)$ process
- ▶ Forecasting for the $AR(p)$ process

Introduction

During previous lectures, we have considered model types for time series and how to fit such models.

An important aspect of time series analysis is predicting the behaviour of a time series at future points in time.

We refer to prediction as **forecasting**. Typically, we are interested in making short-term forecasts for a time series, that is, predicting future behaviour over a short period of time.

Often, it would not be too sensible to attempt to make long-term forecasts because there will usually be a great amount of uncertainty with regard to the behaviour of a series at a distant future point in time.

In this course, we shall focus only on short-term forecasting.

Forecasting – Issues

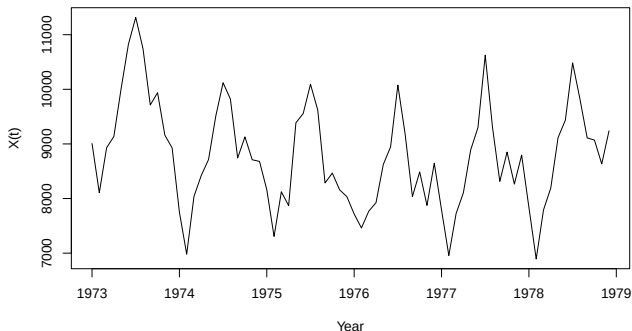
There are challenges when making forecasts:

1. An incorrect model may have been fitted (e.g. an $ARMA(p, q)$ model with the wrong values for p and q). Naturally, forecasts from an incorrect model are unlikely to be reliable.
2. If a model appears to fit some data well, each parameter estimate will have associated with it uncertainty (reflected in the standard errors of parameter estimates). This uncertainty should be used when making predictions.
3. Suppose that at time $t = n + k$ we would like to predict the value of the series X_t , having observed data up to time $t = n$ (where $k > 0$).

On paper, this would involve using future values of white noise $Z_{n+1}, \dots, Z_{n+k}$, but these are unknown!

Forecasting – Example

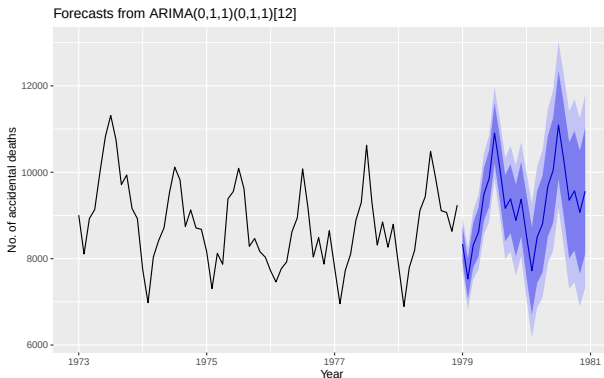
To give an idea of our aims when short-term forecasting, recall from Lecture 10 the data on the monthly number of US accidental deaths from 1973 to 1978, for which a time plot is shown below.



Suppose that we would like to forecast values from this series for the years 1979 and 1980.

Forecasting – Example

The plot below shows the observed series (1973–1978) together with some predicted values for 1979 and 1980. The shaded areas give uncertainty bounds (80% and 95%) around the forecasted estimates.



The above plot was obtained using the forecast package in R.

Prediction - Background

Before making predictions based on time series models, we consider the following statement, which will be useful later on.

Statement 11.1

If Y is a random variable with $\mathbb{E}(Y) = \mu$ and $\mathbb{E}(Y^2) < \infty$ then, for any constant α , the expectation $\mathbb{E}[(Y - \alpha)^2]$ is minimised when $\alpha = \mu$. This is because

$$\begin{aligned}\mathbb{E}[(Y - \alpha)^2] &= \mathbb{E}[(Y - \mu + \mu - \alpha)^2] \\ &= \mathbb{E}[(Y - \mu)^2] + (\alpha - \mu)^2.\end{aligned}$$

Since each term on the right hand side involves a square, each will be ≥ 0 and, clearly, $(\alpha - \mu)^2$ is minimised ($= 0$) where $\alpha = \mu$.

Forecasting - Background

Suppose that at time n we have observed $X_1, \dots, X_n$ and we denote these observations $\mathbf{x} = (x_1, \dots, x_n)^\top$.

Our aim is to predict X_{n+k} for some $k > 0$ (i.e. the value of the time series at time $n + k$).

We would like to make this prediction by making use of the data collected up to time n (as, clearly, these data provide information about the past behaviour of the time series to be used to predict the future).

To encapsulate the information from the data up to time n , we can define a function F whose argument is $\mathbf{x} = (x_1, \dots, x_n)^\top$.

This gives rise to the question: 'what type of function, F , should we choose?'

Forecasting - Background

One way of defining F , which we shall use, is to make use of Statement 11.1 and to choose a function F that minimises the **mean square prediction error**.

That is, we choose $F(\mathbf{x})$ such that

$$\mathbb{E}[(X_{n+k} - F(\mathbf{x}))^2]$$

is minimised.

Using Statement 11.1, (setting $Y = X_{n+k} \mid \mathbf{x}$) we choose

$$F(\mathbf{x}) = \mathbb{E}(X_{n+k} \mid x_1, \dots, x_n)$$

and we restrict $\mathbb{E}(X_{n+k} \mid x_1, \dots, x_n)$ to be linear.

Forecasting

Suppose that $\hat{X}_{n+k}(n)$ is the predicted (forecasted) value of X_{n+k} , having observed the time series up to time n .

Since we have restricted $F(\mathbf{x})$ to be linear, we define

$$\begin{aligned}\hat{X}_{n+k}(n) &= \sum_{j=0}^{\infty} d_j(k) X_{n-j} \\ &= d_0(k) X_n + d_1(k) X_{n-1} + d_2(k) X_{n-2} + \dots\end{aligned}$$

and we seek the coefficients $d_j(k)$ that minimise the mean square prediction error

$$\mathbb{E} \left[\left(X_{n+k} - \sum_{j=0}^{\infty} d_j(k) X_{n-j} \right)^2 \right].$$

We'll now consider this approach for some of our familiar time series models.

Forecasting for ARMA processes

We assume that $\{X_t: t \in \mathbb{Z}\}$ is a zero mean ARMA process and, therefore, X_t has the following $MA(\infty)$ representation

$$X_t = \sum_{i=0}^{\infty} b_i Z_{t-i}$$

with $b_0 = 1$, $\sum_{i=0}^{\infty} b_i^2 < \infty$ and $\text{Var}(Z_t) = \sigma_z^2$.

The linear predictor for the observation $\hat{X}_{n+k}$ can then be written

$$\begin{aligned}\hat{X}_{n+k}(n) &= \sum_{j=0}^{\infty} d_j(k) X_{n-j} \\ &= \sum_{j=0}^{\infty} c_j(k) Z_{n-j}\end{aligned}$$

for some other coefficients $c_j(k)$ since the X_t terms are all functions of Z_t terms.

Forecasting for ARMA processes

Then the mean square prediction error is

$$\begin{aligned}\mathbb{E} \left[\left(X_{n+k} - \hat{X}_{n+k}(n) \right)^2 \right] &= \mathbb{E} \left[\left(\sum_{j=0}^{\infty} b_j Z_{n+k-j} - \sum_{j=0}^{\infty} c_j(k) Z_{n-j} \right)^2 \right] \\ &= \mathbb{E} \left[\left(\sum_{j=0}^{k-1} b_j Z_{n+k-j} + \sum_{j=0}^{\infty} (b_{j+k} - c_j(k)) Z_{n-j} \right)^2 \right]\end{aligned}$$

Now, since the Z_t variables are white noise, it follows that $\mathbb{E}(Z_i Z_j) = 0$ when $i \neq j$.

Then, expanding the above expectation, we see that...

Forecasting for ARMA processes

Forecasting for ARMA processes

We see that the expectation will be minimised where

$$\begin{aligned} b_{j+k} - c_j(k) &= 0 \quad \text{for all } j \\ \implies c_j(k) &= b_{j+k}. \end{aligned}$$

Hence

$$\begin{aligned} \hat{X}_{n+k}(n) &= \sum_{j=0}^{\infty} c_j(k) Z_{n-j} \\ &= \sum_{j=0}^{\infty} b_{j+k} Z_{n-j} \end{aligned}$$

Forecasting for ARMA processes

We recall that

$$X_t = \sum_{j=0}^{\infty} b_j Z_{t-j}$$

and substitution of $t = n + k$ into the above yields

$$\begin{aligned} X_{n+k} &= \sum_{j=0}^{\infty} b_j Z_{n+k-j} \\ &= b_0 Z_{n+k} + b_1 Z_{n+k-1} + \dots + b_k Z_n + \dots \end{aligned}$$

Forecasting for ARMA processes

Then, since the mean square prediction error is

$$\begin{aligned}\mathbb{E} \left[\left(X_{n+k} - \hat{X}_{n+k}(n) \right)^2 \right] &= \mathbb{E} \left[\left(X_{n+k} - \sum_{j=0}^{\infty} b_{j+k} Z_{n-j} \right)^2 \right] \\&= \mathbb{E} \left[\left(\sum_{j=0}^{\infty} b_j Z_{n+k-j} - \sum_{j=0}^{\infty} b_{j+k} Z_{n-j} \right)^2 \right] \\&= \mathbb{E} \left[\left(\sum_{j=0}^{k-1} b_j Z_{n+k-j} \right)^2 \right] \\&= \sum_{j=0}^{k-1} b_j^2 \mathbb{E}(Z_{n+k-j}^2) + \sum_{i \neq j} b_i b_j \mathbb{E}(Z_{n+k-i} Z_{n+k-j})\end{aligned}$$

Forecasting for ARMA processes

Since the Z_t variables (white noise) are independent

$$\sum_{i \neq j} b_i b_j \mathbb{E}(Z_{n+k-i} Z_{n+k-j}) = 0$$

and it follows that the mean square prediction error is

$$\begin{aligned} \mathbb{E} \left[\left(X_{n+k} - \hat{X}_{n+k}(n) \right)^2 \right] &= \sum_{j=0}^{k-1} b_j^2 \mathbb{E}(Z_{n+k-j}^2) \\ &= \sigma_z^2 \sum_{j=0}^{k-1} b_j^2. \end{aligned}$$

Hence, we would like to choose $b_1, \dots, b_{k-1}$ (remembering $b_0 = 1$) so that the above expression is minimised.

Forecasting for ARMA processes

Having obtained, the values of $b_1, \dots, b_{k-1}$, we need to find the values of $d_j(k)$ to make the prediction

$$\hat{X}_{n+k}(n) = \sum_{j=0}^{\infty} d_j(k) X_{n-j}.$$

To do this, we define the following functions

$$B(\xi) = \sum_{m=0}^{\infty} b_m \xi^m$$

$$C_k(\xi) = \sum_{j=0}^{\infty} c_j(k) \xi^j = \sum_{j=0}^{\infty} b_{j+k} \xi^j$$

$$D_k(\xi) = \sum_{j=0}^{\infty} d_j(k) \xi^j.$$

Forecasting for ARMA processes

Each of the functions above is a generating function and the b_j coefficients are known because these are coefficients from the $MA(\infty)$ representation of the process.

It can be shown that

$$C_k(\xi) = B(\xi)D_k(\xi).$$

Then

$$D_k(\xi) = \frac{C_k(\xi)}{B(\xi)} = \frac{\sum_{j=0}^{\infty} b_{j+k}\xi^j}{\sum_{m=0}^{\infty} b_m\xi^m}.$$

The above may be expanded out as a polynomial with terms $\xi, \xi^2, \dots$ and then the coefficient of ξ^j in $D_k(\xi)$ may be evaluated as $d_j(k)$.

Forecasting for ARMA processes

Once the coefficients $d_j(k)$ have been determined, the forecast at k steps ahead in time is given by

$$\hat{X}_{n+k}(n) = \sum_{j=0}^{\infty} d_j(k) X_{n-j}.$$

Although this approach may appear complicated, we will use the result for specific ARMA models to derive the form of the predicted value at $t = n + k$.

Forecasting for ARMA processes

At this point, we make some comments. Suppose that we are at time point $t = n$ in the series and subsequent time points are 'the future'

1. We know that

$$\begin{aligned} X_{n+k} &= \sum_{j=0}^{\infty} b_j Z_{n+k-j} \\ &= \underbrace{(b_0 Z_{n+k} + \dots + b_{k-1} Z_{n+1})}_{\text{'the future'}} + \underbrace{(b_k Z_n + b_{k+1} Z_{n-1} + \dots)}_{\text{'the past'}} \\ &= \sum_{j=0}^{k-1} b_j Z_{n+k-j} + \sum_{j=0}^{\infty} b_{j+k} Z_{n-j} \\ &= \text{Prediction error} + \hat{X}_{n+k}(n) \end{aligned}$$

since $\hat{X}_{n+k}(n) = \sum_{j=0}^{\infty} b_{j+k} Z_{n-j}$.

2. The mean square prediction error, σ_k^2 is

$$\sigma_k^2 = \sigma_z^2 \sum_{j=0}^{k-1} b_j^2.$$

We note that, as $k \rightarrow \infty$, $\sigma_k^2 \rightarrow \sigma_z^2 \sum_{j=0}^{\infty} b_j^2 = \text{Var}(X_t)$

This result is intuitive since we should expect the variance of our prediction to increase when we attempt to make longer term forecasts (i.e. as k increases).

3. If $k = 1$, the prediction error is equal to Z_{n+1} , which makes sense!

Example: $AR(2)$ process

We will now consider how to make predictions for an $AR(2)$ process.

Recall that the $AR(2)$ process is written

$$X_t - \phi_1 X_{t-1} - \phi_2 X_{t-2} = Z_t.$$

We derive the $MA(\infty)$ representation of the $AR(2)$ process as

$$(1 - \lambda_1 B)(1 - \lambda_2 B)X_t = Z_t$$

and assume that we have two, distinct, real roots (i.e. $\frac{1}{\lambda_1} \neq \frac{1}{\lambda_2}$).

Example: $AR(2)$ process

Then

$$\begin{aligned}(1 - \lambda_1 B)(1 - \lambda_2 B)X_t &= Z_t \\ \implies X_t &= (1 - \lambda_1 B)^{-1}(1 - \lambda_2 B)^{-1}Z_t \\ &= \sum_{j=0}^{\infty} \underbrace{\left(\frac{\lambda_1^{j+1} - \lambda_2^{j+1}}{\lambda_1 - \lambda_2} \right)}_{= b_j} Z_{t-j}.\end{aligned}$$

The above is a linear function of $Z_t, Z_{t-1}, \dots$

Example: $AR(2)$ process

Using results derived earlier, the predictor $\hat{X}_{n+k}(n)$ is

$$\begin{aligned}\hat{X}_{n+k}(n) &= \sum_{j=0}^{\infty} b_{j+k} Z_{n-j} \\ &= \sum_{j=0}^{\infty} \left(\frac{\lambda_1^{j+k+1} - \lambda_2^{j+k+1}}{\lambda_1 - \lambda_2} \right) Z_{n-j} \\ &= \frac{\lambda_1^{k+1}}{\lambda_1 - \lambda_2} \sum_{j=0}^{\infty} \lambda_1^j Z_{n-j} - \frac{\lambda_2^{k+1}}{\lambda_1 - \lambda_2} \sum_{j=0}^{\infty} \lambda_2^j Z_{n-j} \quad (1).\end{aligned}$$

However, we require

$$\hat{X}_{n+k}(n) = \sum_{j=0}^{\infty} d_j(k) X_{n-j}.$$

i.e. $\hat{X}_{n+k}(n)$ as a function of $X_n, X_{n-1}, \dots$ and not $Z_n, Z_{n-1}, \dots$

Example: $AR(2)$ process

Note that

$$(1 - \lambda_1 B)(1 - \lambda_2 B)X_n = Z_n$$

$$\implies (1 - \lambda_1 B)X_n = (1 - \lambda_2 B)^{-1}Z_n = \sum_{j=0}^{\infty} \lambda_2^j Z_{n-j} \quad (2)$$

$$\implies (1 - \lambda_2 B)X_n = (1 - \lambda_1 B)^{-1}Z_n = \sum_{j=0}^{\infty} \lambda_1^j Z_{n-j} \quad (3)$$

Substitution of (2) and (3) into (1) yields...

Example: $AR(2)$ process

$$\begin{aligned}\hat{X}_{n+k}(n) &= \frac{\lambda_1^{k+1}}{\lambda_1 - \lambda_2} (1 - \lambda_2 B) X_n - \frac{\lambda_2^{k+1}}{\lambda_1 - \lambda_2} (1 - \lambda_1 B) X_n \\ &= \frac{\lambda_1^{k+1}}{\lambda_1 - \lambda_2} (X_n - \lambda_2 X_{n-1}) - \frac{\lambda_2^{k+1}}{\lambda_1 - \lambda_2} (X_n - \lambda_1 X_{n-1}) \\ &= \frac{1}{\lambda_1 - \lambda_2} \left\{ (\lambda_1^{k+1} - \lambda_2^{k+1}) X_n - (\lambda_2 \lambda_1^{k+1} - \lambda_1 \lambda_2^{k+1}) X_{n-1} \right\}.\end{aligned}$$

Note that the above expression for the predicted value $\hat{X}_{n+k}(n)$ involves only λ_1, λ_2 and the current and previous values of the process (X_n and X_{n-1}).

Example 11.1

Consider the (weakly) stationary $AR(2)$ process

$$X_t = \frac{3}{4}X_{t-1} - \frac{1}{8}X_{t-2} + Z_t.$$

If the process is observed up to time n , what is the predicted value of this process at times

(a) $n + 1$?

(b) $n + 2$?

Example 11.1

Example 11.2: $AR(1)$ process

Example 11.2: $AR(1)$ process

Forecasting: $AR(p)$ process

Recall that the $AR(p)$ process is written

$$\phi(B)X_t = Z_t$$

where

$$\phi(B) = 1 - \phi_1 B - \phi_2 B^2 - \dots - \phi_p B^p.$$

We know that

$$X_{n+k} = \hat{X}_{n+k}(n) + \text{Prediction Error}$$

and note that the prediction error is a function of $Z_{n+1}, Z_{n+2}, \dots$

Forecasting: $AR(p)$ process

The true, but unknown, value of the process at time $n + k$ is given by

$$\begin{aligned} X_{n+k} &= \sum_{j=0}^{\infty} b_j Z_{n+k-j} \\ &= \left\{ \sum_{j=0}^{k-1} b_j Z_{n+k-j} \right\} + \left\{ \sum_{j=k}^{\infty} b_j Z_{n+k-j} \right\} \\ &= \text{Future Error} + \hat{X}_{n+k}(n). \end{aligned}$$

Forecasting: $AR(p)$ process

We note that

$$\sum_{j=k}^{\infty} b_j Z_{n+k-j} = \sum_{j=0}^{\infty} b_j [Z_{n+k-j}]$$

where

$$[Z_{n+k-j}] = \begin{cases} 0 & j < k \\ Z_{n+k-j} & j \geq k. \end{cases}$$

Forecasting: $AR(p)$ process

We see that

$$\hat{X}_{n+k}(n) = \sum_{j=0}^{\infty} b_j [Z_{n+k-j}]$$

This is similar to the 'true' $MA(\infty)$ form of X_{n+k} , which is

$$X_{n+k} = \sum_{j=0}^{\infty} b_j Z_{n+k-j}.$$

So to obtain the predicted value of X_{n+k} , we take the $MA(\infty)$ form of X_{n+k} but replace the terms Z_{n+k-j} with $[Z_{n+k-j}]$.

Forecasting: $AR(p)$ process

It follows that

$$\phi(B)\hat{X}_{n+k}(n) = [Z_{n+k}].$$

with

$$[Z_{n+k}] = \begin{cases} 0 & k > 0 \\ Z_{n+k} & k \leq 0. \end{cases}$$

Forecasting: $AR(p)$ process

Hence, where $k = 1$, we see that

$$\begin{aligned}\phi(B)\hat{X}_{n+1}(n) &= [Z_{n+1}] = 0 \\ \implies \hat{X}_{n+1}(n) - \phi_1\hat{X}_n(n) - \dots - \phi_p\hat{X}_{n-p+1}(n) &= 0\end{aligned}$$

Rearranging, we deduce that

$$\hat{X}_{n+1}(n) = \phi_1\hat{X}_n(n) + \dots + \phi_p\hat{X}_{n-p+1}(n).$$

But, $X_n, \dots, X_{n-p+1}$ have already been observed!

Hence, naturally, we set

$$(\hat{X}_n(n), \dots, \hat{X}_{n-p+1}(n)) = (X_n, \dots, X_{n-p+1}).$$

Then

$$\hat{X}_{n+1}(n) = \phi_1X_n + \dots + \phi_pX_{n-p+1}.$$

Forecasting: $AR(p)$ process

If $k = 2$ we know that

$$\hat{X}_{n+2}(n) = \phi_1 \hat{X}_{n+1}(n) + \phi_2 \hat{X}_n(n) + \dots + \phi_p \hat{X}_{n-p+2}(n).$$

Here, we'd compute $\hat{X}_{n+1}(n)$ as shown on the previous slide and replace $\hat{X}_n(n), \dots, \hat{X}_{n-p+2}(n)$ with $X_n, \dots, X_{n-p+2}$ because those values have been observed.

In general, for an $AR(p)$ model

$$\hat{X}_{n+k}(n) = \phi_1 \hat{X}_{n+k-1}(n) + \phi_2 \hat{X}_{n+k-2}(n) + \dots + \phi_p \hat{X}_{n+k-p}(n)$$

and $\hat{X}_j(n)$ is replaced with X_j if $j \leq n$.

Forecasting: $AR(p)$ process

Having seen the theory for forecasting from $AR(p)$ processes, we consider **how** we'd make predictions **in practice**, having observed data $x_1, \dots, x_n$.

1. First, we transform the observations by calculating the sample mean, $\bar{x}$, and defining the transformed series $y_t = x_t - \bar{x}$.
2. Using our knowledge of $AR(p)$ process estimation (from previous lectures) we estimate the coefficients $\phi_1, \dots, \phi_p$, defining the estimates $\hat{\phi}_1, \dots, \hat{\phi}_p$.
3. Using the estimates from 2., predictions are made for the transformed series. The sample mean, $\bar{x}$, is then added to each predicted value to report predictions on the scale of the original series.

Forecasting: $AR(p)$ process

In addition to estimating values at future points in time, we'd also like to estimate the **uncertainty** when making forecasts.

For an $AR(p)$ process, we know that

$$\begin{aligned}\sigma_k^2 &= \sigma_z^2 \sum_{j=0}^{k-1} b_j^2 \\ &\approx \hat{\sigma}_z^2 \sum_{j=0}^{k-1} \hat{b}_j^2 = \hat{\sigma}_k^2.\end{aligned}$$

An approximate 95% confidence interval for the prediction, $\hat{X}_{n+k}(n)$ - from the $AR(p)$ process - is given by

$$\left(\hat{X}_{n+k}(n) - 2\hat{\sigma}_k, \hat{X}_{n+k}(n) + 2\hat{\sigma}_k \right).$$

Lecture 11: Learning Outcomes

- ▶ Understand what is meant by forecasting and be aware of some of the issues and challenges when attempting to make predictions using time series data.
- ▶ Know what is meant by the mean square prediction error and how this measure may be used in forecasting.
- ▶ Understand how predictions are made from ARMA processes by minimising the mean square prediction error.
- ▶ Understand the presented examples for the $AR(2)$, $AR(1)$ and $AR(p)$ processes.
- ▶ Understand how predictions may be made, in practice, for $AR(p)$ processes.